What is Computer Graphics? List out the application of Computer graphics. Differentiate between CAD and CAM.

What is Computer Graphics?

Computer Graphics is the field of computer science that focuses on generating, manipulating, and storing images and geometric information via digital methods. It involves the use of specialized software and hardware to convert data into visual representations, ranging from simple 2D line drawings to complex 3D animations and virtual reality environments.

At its core, computer graphics is about the interaction between computers and visual data. This includes:

- ✓ **Modeling:** Defining the shape and characteristics of objects.
- ✓ **Rendering:** Converting the model into an image by calculating light, shadow, and texture.
- ✓ **Animation:** Creating the illusion of motion by displaying images in rapid succession.

Applications of Computer Graphics

Computer graphics are used across almost every modern industry. Some of the most common applications include:

- ✓ **Computer-Aided Design (CAD):** Used in engineering, architecture, and manufacturing to create precise drawings and models of buildings, bridges, and mechanical parts.
- ✓ **Entertainment:** The foundation of movies, video games, and television special effects (VFX).
- ✓ **Medical Imaging:** Used to visualize internal organs and structures through techniques like CT scans, MRIs, and 3D ultrasound.
- ✓ **Education and Training:** Creating flight simulators for pilots, surgical simulations for doctors, and interactive educational software.
- ✓ **Data Visualization:** Converting complex sets of data into charts, graphs, and heat maps to make information easier to understand.

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- ✓ **Graphic Design:** Creating advertisements, logos, and digital art for print and web media.
- ✓ **Virtual and Augmented Reality (VR/AR):** Providing immersive environments for gaming, remote work, and industrial training.

Difference Between CAD and CAM

Feature	Computer-Aided Design (CAD)	Computer-Aided Manufacturing (CAM)
Primary Goal	To **design** and document the physical characteristics of a product.	To **control** and manage the machinery used to produce the product.
Focus	Geometric modeling, aesthetics, and technical specifications.	Toolpaths, machining speeds, and production efficiency.
Output	2D drawings or 3D digital models (e.g., .STL, .DWG).	G-code or instructions that a machine (like a CNC) can follow.
User Role	Used by engineers, architects, and designers to create concepts.	Used by machinists and manufacturing engineers to execute designs.
Process Order	Occurs at the **start** of the product development cycle.	Occurs **after** the design is finalized to facilitate production.

Explain different types of touch panels. Different between raster and Vector scan display architectures.

Types of Touch Panels

Touch panels are sensors that allow users to interact with a computer system by touching the screen. There are several technologies used to achieve this:

1. Resistive Touch Panels

These consist of two thin, flexible metallic layers separated by a narrow gap. When you press the screen, the layers touch, completing an electrical circuit.

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- ✓ **Pros:** Can be used with fingers, gloves, or a stylus; very affordable.
- ✓ **Cons:** Not very clear (lower image quality); no multi-touch support; easily damaged by sharp objects.

2. Capacitive Touch Panels

These panels use a layer of glass coated with a transparent conductor. Since the human body conducts electricity, touching the screen pulls a small amount of charge to the point of contact.

- ✓ **Surface Capacitive:** Sensors at the corners detect the touch.
- ✓ **Projected Capacitive (P-Cap):** Uses a grid of rows and columns. This is the technology used in **smartphones**.

- ✓ **Pros:** Highly durable; excellent clarity; supports multi-touch (pinching/zooming).
- ✓ **Cons:** Generally requires a conductive input (bare finger or special stylus); won't work with regular gloves.

3. Infrared (IR) Touch Panels

These use a frame of IR LEDs and photodetectors around the edges of the screen. When you touch the screen, you block the invisible light beams.

- ✓ **Pros:** Excellent image clarity (no extra layers over the screen); highly durable.
- ✓ **Cons:** Dust or debris on the frame can interfere with the sensors.

4. Surface Acoustic Wave (SAW)

This technology uses ultrasonic waves that pass over the screen. Touching the screen absorbs a portion of the wave.

- ✓ **Pros:** High clarity and durability.
- ✓ **Cons:** Can be triggered by contaminants like dirt or moisture.

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Raster Scan vs. Vector Scan Display

Feature	**Raster Scan Display**	**Vector Scan (Random Scan) Display**
Electron Beam Path	The beam sweeps across the entire screen, row by row (top to bottom).	The beam is directed only to the parts of the screen where the image is to be drawn.
Basic Building Block	Uses **Pixels** (dots).	Uses **Line Commands** (mathematical vectors).
Image Storage	Picture definition is stored in a **Frame Buffer** as intensity values for every pixel.	Picture definition is stored as a set of line-drawing commands in a **Display File**.
Resolution	Limited by the number of pixels in the frame buffer.	Higher resolution; produces very smooth lines.
Color/Shading	Excellent for displaying realistic images with complex shading and colors.	Difficult to produce realistic shaded images; best for wireframe drawings.
Refresh Rate	Constant, regardless of image complexity (typically 60Hz - 120Hz).	Depends on the number of lines to be drawn; complex images can cause flickering.

What is scalding? Write bresenham's line drawing algorithm.

What is Scaling?

In computer graphics, Scaling is a transformation process used to change the size of an object. This is done by multiplying the coordinate values (x, y) of each vertex of the object by a Scaling Factor (S_x for the x-direction and S_y for the y-direction).

- ✓ If the scaling factor is greater than 1, the object expands (gets larger).
- ✓ If it is between 0 and 1, the object compresses (gets smaller).
- ✓ If the scaling factor is 1, the object remains the same size.
- ✓ The transformation can be represented by the following equations:

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$x' = x \cdot S_x$
$y' = y \cdot S_y$

Bresenham's Line Drawing Algorithm

Bresenham's algorithm is an efficient method for determining which points on a 2D raster grid should be plotted to form a close approximation of a straight line. Unlike the DDA algorithm, it uses integer addition, subtraction, and bit shifting, avoiding expensive floating-point operations.

The Algorithm Steps (for m < 1):

Input the two line endpoints (x_0, y_0) and (x_1, y_1).

Calculate the constants:

$\Delta x = |x_1 - x_0|$

$\Delta y = |y_1 - y_0|$

Calculate the initial decision parameter:

$P_0 = 2\Delta y - \Delta x$
4 At each x_k along the line, starting at k=0, perform the following test:

- If $P_k < 0$, the next point to plot is $(x_k + 1, y_k)$ and:
- $P_{k+1} = P_K + 2\Delta y$
- Otherwise, the next point to plot is $(x_k + 1, y_k + 1)$and:
- $P_{k+1} = P_k + 2\Delta y - 2\Delta x$

Repeat step 4 Δx times until the destination point is reached.

Given a triangle with corner coordinates(0,0),(2,0) and (3,2) perform 2D rotation by 90^0 anticlockwise and find out new coordinate. Write DDA line drawing algorithm.

2D Rotation of the Triangle

To rotate a point (x, y) by an angle θ anticlockwise about the origin (0, 0), we use the following transformation equations:

$x' = x \cos\theta - y \sin\theta$
$y' = x \sin\theta + y \cos\theta$
For a **90^0 anticlockwise rotation**:

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$\cos(90^0) = 0$

$\sin(90^0 circ) = 1$

Substituting these values, the equations become:

x' = -y
y' = x

Calculations for the Triangle:

Given the vertices of the triangle: A(0, 0),B(2, 0), and C(3, 2).

For Point A(0, 0):

x' = -0 = 0

y' = 0

New Coordinate A' = (0, 0)

For Point B(2, 0):

x' = -0 = 0

y' = 2

New Coordinate B' = (0, 2)

For Point C(3, 2):

x' = -2

y' = 3

New Coordinate C' = (-2, 3)

Result: The new coordinates after a $90^\circ anticlockwise rotation are **(0, 0), (0, 2), and (-2, 3)**

2. DDA Line Drawing Algorithm

The **Digital Differential Analyzer (DDA)** algorithm is a scan-conversion line algorithm based on calculating either Δx or Δy. It uses floating-point arithmetic to determine the next pixel position along a line path.

Algorithm Steps:

Input the two endpoints of the line: (x_1, y_1) and (x_2, y_2).

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Calculate the differences:

$dx = x_2 - x_1$

$dy = y_2 - y_1$

Determine the number of steps:

If |dx| > |dy|, then steps = |dx|

Else, steps = |dy|

Calculate the increment in x and y for each step:

xinc = dx /steps

yinc = dy /steps

Initialize the starting point:

x = x1

y = y1

Loop for i = 0 to steps:

Plot the point (round(x),(round(y)) on the screen.

Update the coordinates:

x = x + xinc

y = y + yinc

Write the matrix notation for shearing. Explain different steps involved in 3D viewing pipeline.

1. Matrix Notation for Shearing

Shearing is a transformation that slants the shape of an object. In 2D, we can have shearing in the X-direction, Y-direction, or both.

2. 3D Viewing Pipeline

The **3D Viewing Pipeline** is the series of steps used to convert a 3D scene (defined in world coordinates) into a 2D image that can be displayed on a monitor.

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Step 1: Object Coordinates to World Coordinates

Objects are initially defined in their own local coordinate systems (Modeling Coordinates). These are transformed into a single global **World Coordinate System** using translation, rotation, and scaling.

Step 2: World Coordinates to Viewing Coordinates

We define a "Camera" or "Eye" position. The entire world is then transformed so that the camera is at the origin $(0,0,0)$ and is looking along a specific axis (usually the Z-axis). This is called **Viewing Transformation**.

Step 3: Viewing Coordinates to Projection Coordinates

Since a monitor is 2D, we must project the 3D scene onto a 2D plane.

- ✓ **Parallel Projection:** Keeps the relative proportions (used in engineering/CAD).
- ✓ **Perspective Projection:** Makes distant objects appear smaller (realistic view).

Step 4: Normalized Coordinates (Clipping)

The viewing volume (the part of the space visible to the camera) is converted into a **Normalized Device Coordinate (NDC)** system (usually ranging from -1 to 1). At this stage, any parts of the objects outside the view volume are "clipped" or removed.

Step 5: Device Coordinates (Viewport Transformation)

Finally, the normalized coordinates are mapped to the actual **Device Coordinates** (pixel coordinates) of the specific screen or window where the image will be displayed.

Define perspective projection. Differentiate between object space and image space method in hidden surface removal technique.

Perspective Projection

Perspective Projection is a 3D transformation technique used to represent a 3D object on a 2D surface in a way that mimics how the human eye or a camera perceives the world.

The most defining characteristic of perspective projection is that **objects farther away from the viewer appear smaller** than objects of the same size that are closer. All parallel lines that are not parallel to the view plane appear to converge at a single point called the **Vanishing Point**.

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Key Properties:

- ✓ **Center of Projection:** The point where the "eye" or camera is located.
- ✓ **Vanishing Point:** The point where parallel lines seem to meet in the distance.
- ✓ **Foreshortening:** The phenomenon where the size of the projected object varies inversely with its distance from the center of projection.

Object Space vs. Image Space Methods

In hidden surface removal (visible surface detection), we need to decide which parts of a 3D scene are visible and which are blocked by other objects. There are two primary approaches to solving this:

Feature	Object Space Method	Image Space Method
Basic Concept	Compares objects and parts of objects directly with each other to determine visibility.	Determines visibility at each **pixel** position on the projection plane.
Logic	"Which object is in front of another?"	"Which object is visible at this specific pixel (x, y)?"
Precision	Very high; calculations are done using the original coordinate data.	Limited by the resolution of the display (frame buffer).
Complexity	Increases with the number of objects ($O(n^2)$ complexity).	Increases with the number of pixels and objects (O(np) complexity).
Examples	Back-Face Detection, Area Sub-division (Warnock's Algorithm).	**Z-Buffer (Depth Buffer)**, Scan-line Method.
Common Use	Best for scenes with a few complex objects or for high-precision vector output.	Most common in modern 3D graphics and gaming because it scales better with complex scenes.

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What is parallel projection? Explain constant shading and phong shading model.

To ensure that information is exchanged safely over a network, a system must satisfy specific security requirements. In the field of cyber security and data communication, these are often referred to as the CIA Triad, with a few additional pillars added for modern networks.

Here are the five essential properties of secure communication:

1. Confidentiality

Confidentiality ensures that only the intended recipient can read the message. Even if a hacker intercepts the data, they should not be able to understand its contents.

- ✓ **How it's achieved:** Primarily through **Encryption** (converting plain text into cipher text using algorithms like AES or RSA).
- ✓ **Example:** When you send a private message, only you and your friend have the "keys" to read it.

2. Integrity

Integrity guarantees that the message has not been altered, deleted, or tampered with during transmission. The receiver must be sure that the data they see is exactly what the sender sent.

- ✓ **How it's achieved:** Using **Hash functions** (like SHA-256) or **Digital Signatures**. If even one bit of the data changes, the "hash" value will change, alerting the receiver.
- ✓ **Example:** Ensuring that a bank transfer for **Rs. 1,000** doesn't get changed to **Rs. 10,000** by a middleman.

3. Authentication

Authentication is the process of verifying the identity of the parties involved in the communication. The receiver needs to be sure the message really came from the claimed sender, and the sender needs to know they are talking to the correct receiver.

- ✓ **How it's achieved:** Through **Passwords**, **Digital Certificates**, or **Biometrics**.
- ✓ **Example:** Logging into your e-Governance portal requires a username and password to prove it's really you.

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4. Availability

Availability ensures that the communication system and the data are accessible to authorized users whenever they are needed. A secure system is useless if it is constantly crashing or blocked by attacks.

- ✓ **How it's achieved:** Using **Firewalls**, **DDoS protection**, and **Redundant systems** (backup servers).
- ✓ **Example:** Ensuring that a government tax filing website stays online even during high-traffic periods near the deadline.

5. Non-Repudiation

Non-repudiation prevents a sender from later denying that they sent a specific message. Similarly, a receiver cannot deny having received it. It provides legal "proof" of the transaction.

- ✓ **How it's achieved:** Using **Digital Signatures** and **Public Key Infrastructure (PKI)**. Since only the sender has their private key, only they could have signed the message.
- ✓ **Example:** Once you digitally sign a contract or a land registry document, you cannot claim later that "I didn't send that."

What are different graphics file formats? Explain different features of graphics packages.

computer graphics, file formats determine how visual data is stored, while graphics packages provide the tools to create and manipulate that data.

1. Common Graphics File Formats

Graphics files are generally divided into two main categories: **Raster (Bitmap)** and **Vector**.

A. Raster Formats (Pixel-based)

These store images as a grid of colored pixels. They lose quality when enlarged (pixelation).**JPEG/JPG (Joint Photographic Experts Group):** Best for photographs. It uses "lossy" compression, meaning it reduces file size by removing some data.

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- ✓ **PNG (Portable Network Graphics):** Supports **transparency**. It uses "lossless" compression, making it ideal for web logos and icons.
- ✓ **GIF (Graphics Interchange Format):** Limited to 256 colors. It is primarily used for simple **animations**.
- ✓ **TIFF (Tagged Image File Format):** High-quality, large files used in professional printing and scanning.

B. Vector Formats (Path-based)

These store images as mathematical paths (lines and curves). They can be scaled to any size without losing quality.

- ✓ **SVG (Scalable Vector Graphics):** The standard for web-based vector graphics.
- ✓ **AI (Adobe Illustrator):** The industry standard for professional logo design and illustrations.
- ✓ **EPS (Encapsulated PostScript):** A common format used for high-quality printing of vector images.

2. Features of Graphics Packages

A **Graphics Package** is software (like Photoshop, CorelDRAW, or AutoCAD) that allows users to create or edit images. Here are their core features:

I. Drawing and Creation Tools

- ✓ **Primitive Shapes:** Tools to quickly create lines, circles, rectangles, and polygons.
- ✓ **Freehand Drawing:** Tools like the "Pen" or "Brush" that allow for manual artistic creation.
- ✓ **Text Handling:** The ability to add, format, and wrap text around objects.

II. Manipulation and Transformation

- ✓ **Translation (Moving):** Changing the position of an object.
- ✓ **Scaling:** Changing the size of an object.
- ✓ **Rotation:** Turning an object around a pivot point.
- ✓ **Mirroring/Flipping:** Creating a reflected version of the graphic.

III. Editing and Enhancement

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- ✓ **Layering:** The ability to stack different elements on top of each other and edit them independently.
- ✓ **Color Palettes:** Tools to manage RGB, CMYK, and HSL color models.
- ✓ **Filters and Effects:** Pre-programmed algorithms for blurring, sharpening, or adding textures.

IV. Viewing and Navigation

- ✓ **Zooming and Panning:** Navigating through large or detailed designs.
- ✓ **Windowing and Clipping:** Selecting a specific portion of a large image to view or render.

V. Input and Output Management

- ✓ **Device Drivers:** Ensuring the software works with hardware like drawing tablets or high-resolution monitors.
- ✓ **Export Options:** The ability to save the work in various file formats (as discussed above).

What do you mean by Virtual Reality? List out its application. Differentiate between semi-immersive and fully immersive virtual reality

Virtual Reality (VR) is a computer-generated simulation of a three-dimensional environment that can be interacted with in a seemingly real or physical way by a person using special electronic equipment. It tricks the human senses—primarily sight and hearing—into believing they are in a different world.

1. Applications of Virtual Reality

VR is no longer just for gaming; it has revolutionized several professional fields:

- ✧ **Healthcare:** Used for surgical simulations, treating PTSD (Post-Traumatic Stress Disorder), and helping patients manage chronic pain through distraction therapy.

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- **Education & Training:** Allows students to take virtual field trips (e.g., exploring the solar system) or perform dangerous lab experiments in a safe, controlled digital environment.
- **Engineering & Architecture:** Architects use VR to "walk through" a building design before a single brick is laid, helping to identify structural flaws early.
- **Military:** Used for flight simulations, parachute training, and battlefield scenarios to prepare soldiers for high-pressure environments without physical risk.
- **Real Estate:** Potential buyers can take 360-degree virtual tours of properties from anywhere in the world.

2. Semi-Immersive vs. Fully Immersive VR

Feature	Semi-Immersive VR	Fully Immersive VR
Definition	A mix of the real world and a virtual world, often displayed on high-resolution screens.	A complete digital experience where the physical world is entirely shut out.
Equipment	Large projectors, monitors, or high-end flight simulator cockpits.	Head-Mounted Displays (HMDs) like Oculus Rift, noise-canceling headphones, and haptic gloves.
User Awareness	The user remains aware of their physical surroundings (e.g., they can see the floor or the chair they are sitting in).	The user's senses are fully occupied by the simulation; they "lose" sense of the physical room.
Interaction	Interaction is often limited to a joystick, keyboard, or specialized cockpit controls.	Interaction is highly realistic, involving head-tracking, hand gestures, and sometimes full-body movement.
Example	**Flight Simulators** used by pilots for training where they see a digital sky through a cockpit window.	**High-end VR Gaming** or "The Matrix" style simulations where every move you make is mirrored in the digital world.

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Write short notes

a. CRT Monitor

A **CRT (Cathode Ray Tube) Monitor** is a traditional display device that was the standard for computer monitors and televisions for decades before the arrival of LCD and LED screens. It uses a large vacuum tube to display images.

1. How it Works (Mechanism)

- ✓ **Electron Gun:** Located at the back of the monitor, it fires a beam of high-speed electrons.
- ✓ **Deflection System:** Magnetic coils (the "yoke") steer the electron beam horizontally and vertically across the screen.
- ✓ **Phosphor Screen:** The inside of the front glass is coated with **phosphors**. When the electron beam hits these phosphors, they glow to create the image.
- ✓ **Monochrome:** Uses one type of phosphor (usually green or white).
- ✓ **Color:** Uses three types of phosphors (**Red, Green, and Blue - RGB**) arranged in dots or stripes.

2. Key Components

- ✧ **cathode:** The source of the electrons.
- ✧ **Anode:** Accelerates the electrons toward the screen.
- ✧ **Shadow Mask/Aperture Grille:** A metal plate with holes that ensures the electron beams hit the correct colored phosphors accurately.
- ✧ **Glass Envelope:** A vacuum-sealed thick glass container.

Advantages

- ✓ **Color Accuracy:** CRTs are known for excellent color reproduction and deep black levels.
- ✓ **High Contrast:** They provide a very high contrast ratio compared to early LCDs.
- ✓ **No Native Resolution:** Unlike modern monitors, CRTs can display various resolutions without losing much image quality (no "blurring" at lower resolutions).
- ✓ **Fast Response Time:** Almost zero input lag, which made them popular for early competitive gaming.

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4. Disadvantages

- ✓ **Size and Weight:** They are very bulky, heavy, and take up significant desk space.
- ✓ **Power Consumption:** They consume a lot of electricity and generate a significant amount of heat.
- ✓ **Flicker:** Low refresh rates can cause screen flickering, leading to eye strain and headaches.
- ✓ **Radiation & X-rays:** Though shielded, they emit small amounts of electromagnetic radiation.
- ✓ **Distortion:** Images at the corners can sometimes appear curved or distorted over time.

b. Polygon Tables

In computer graphics, 3D objects are typically represented as a collection of polygons (usually triangles or quadrilaterals). **Polygon Tables** are the data structures used to store the geometric definition of these objects in a structured and efficient way.

Instead of repeating coordinate data for every polygon, these tables use a hierarchical method to save memory and make transformations easier.

Why Use Polygon Tables? (Advantages)

- ✓ **Memory Efficiency:** Coordinates for a single vertex are stored only once, even if that vertex is shared by multiple polygons.
- ✓ **Consistency:** If you move one vertex, all connected edges and faces are updated automatically.
- ✓ **Easy Rendering:** It simplifies the process of calculating surface normals (needed for shading) and identifying which edges are "shared" between faces.
- ✓ **Error Checking:** It helps the graphics engine detect "open" meshes or invalid geometry (like an edge that doesn't connect to anything).

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C. Browser safe colors

Browser Safe Colors (also known as the **Web-Safe Palette**) are a set of 216 colors that were guaranteed to look exactly the same across different web browsers and operating systems (like Windows and Mac) in the early days of the internet.

At that time, most computer monitors could only display **256 colors** (8-bit color). If a website used a color outside of this limited range, the browser would try to "mimic" it using a process called **Dithering** (mixing pixels of different colors), which often made the image look grainy or "noisy."

1. The Math Behind the 216 Colors

The 216 colors were chosen based on a specific mathematical formula involving the **RGB (Red, Green, Blue)** color model.

- ✓ Each of the three primary colors (Red, Green, and Blue) was assigned **6 specific intensity levels**.
- ✓ These levels are: **0, 51, 102, 153, 204, and 255**.
- ✓ In Hexadecimal, these values are: **00, 33, 66, 99, CC, and FF**.

The Calculation: Since there are 6 options for Red, 6 for Green, and 6 for Blue, the total number of combinations is:

$6 \times 6 \times 6 = 216$

(The remaining 40 colors out of the 256 were reserved for the operating system's internal use.)

2. Examples of Browser Safe Hex Codes

A color is "Browser Safe" if its Hex code only uses the pairs **00, 33, 66, 99, CC, or FF**.

- ✓ #000000 (Black) — Safe
- ✓ #FFCC00 (Gold) — Safe
- ✓ #336699 (Blue) — Safe
- ✓ #F1A2C3 — Not Safe (uses values like F1, A2, C3)

3. Why they are less relevant today

Today, modern monitors and smartphones support **True Color (24-bit or 32-bit)**, which can display over **16.7 million colors**.

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- ✓ **No Dithering:** Modern devices can render almost any color perfectly, so the 216-color limit is no longer a technical necessity.
- ✓ **Design Freedom:** Web developers now use the full RGB spectrum to create high-quality gradients and realistic images for modern personal websites and portfolios.

4. Why should you still know them?

- ✓ **Legacy Systems:** Occasionally, you might encounter very old industrial displays or low-power embedded systems (IoT devices) that still use 8-bit color.
- ✓ **Accessibility:** Using high-contrast, simple palettes from this list can sometimes help with basic UI visibility.

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